CSB - AI 3

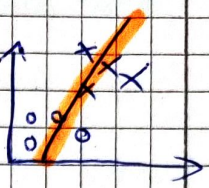
Machine Learning

Normal Programing : Data → Computer → Output
Program →

Machine Learning : Data → Computer → Program
Output →

What is learning?

Binary classification : change coordinates of hyperplane



Linear & non-linear problems

→ Linear Problems :
- Data points can be separated by linear function
- eg : linear regression, SVM

→ Non-linear problems :
- Complex separation function needed
- eg : random forest, NN

Overfitting

Overfitting :

→ Final model perfectly explains the training data, but performs bad on never seen data ↯

Splitting the Data

→ Avoid overfitting by splitting the data into a training and test set

→ Model is trained exclusively on the training data.

→ Subsequently validated on the test data

→ If model is overfitted → bad performance on test data

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Cross-Validation

Problem with test data:

→ if small data set, we waste a large part that one can use for training

→ Model looks different depending on the samples used for the train set

↳ Create multiple splits of the data

K-fold Cross Validation

0) Generate random permutations of sample set
1) Pick a number of folds: k
2) Split the dataset into k equal parts
3) Choose k-1 folds as train sets. The remain fold as test set
4) Repeat 3 until each fold has been used as test data once
5) Identify best model → & Score
6) Repeat with next permutation

Leave-one-out Cross Validation

→ use a single sample as validation set

→ Remain as train data

→ Repeat until every sample has been used exactly once as test data

k-fold with k = # of samples

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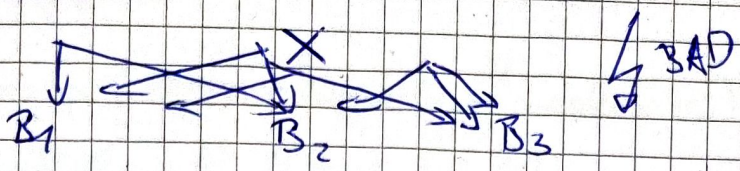
Bootstrap



⇒ Draw bootstrap samples of size n repeatedly from the original data X by simple random sampling with replacement

⇒ leads to lower variance, but has a larger bias than CV

Measuring classification performance

⇒ Assess classifier performance on new, unseen data with known class labels

⇒ Test Data has both positive and negative samples

⇒ classification model generates an output between 0 and 1 for each sample: 1 = positive, 0 = negative } cutoff c

⇒ Compare output to original class labels (ground truth)

TP: positive sample, predicted positive ✓

FN: positive sample, predicted negative ↯

TN: negative sample, predicted negative ✓

FP: negative sample, predicted positive ↯

Sensitivity: $sens = \frac{TP}{TP + FN}$; Specificity: $spec = \frac{TN}{TN + FP}$

Accuracy: $acc = \frac{TP + TN}{n}$

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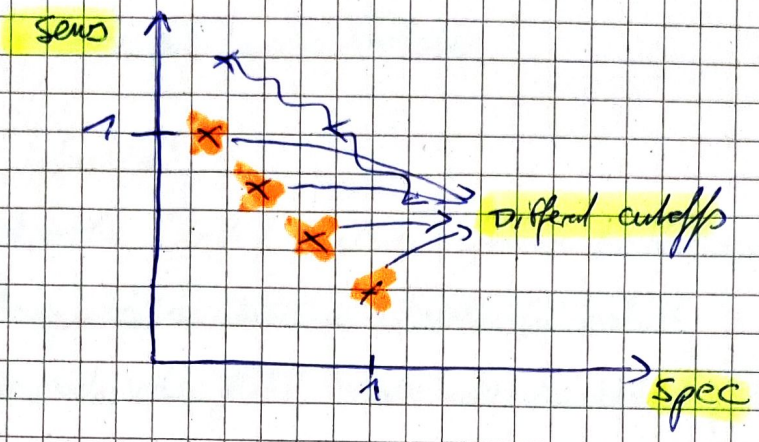
Measuring classification performance

ROC curves (Receiver Operating Characteristics):

- output of the model between [0,1]
- sensitivity and specificity can be computed for various cutoffs



AUC (area under the (ROC) curve):

$$AUC(X) = \int_0^1 ROC(X)$$

, X : classification model

= 1 : perfect classifier

= 0 : no classifier

= 0,5 : toss a coin ≙ random

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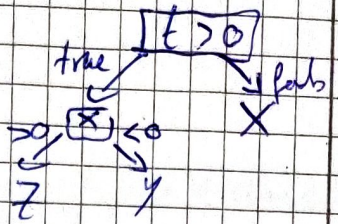
Decision Trees

→ Each Node tests an attribute

→ Each ~~Edge~~ branch corresponds to a value range

→ Each leaf assigns a classification



requires: Attribute-Value pairs,
target function is discrete

Top-Down Induction:

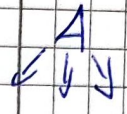
I) A ← the "best" decision attribute for the next node

II) Assign A as decision attribute for node

III) For each value of A, create new descendant of node

IV) Sort training examples to leaf nodes

V) If training examples are perfectly classified STOP
↳ ELSE go to step I)



Quality of a Split ~~and~~ Impurity:

! Construct a tree with leaf nodes that contain cases of a single class only!

→ Node impurity $i(t) = \phi(p_1, \ldots, p_j)$ of node t in split S is any function ϕ where p_j are relative occurrences (frequency of objects of a certain class)

→ requires: → ϕ symmetric for $p_1, \ldots, p_j$
→ ϕ max for even distribution
→ ϕ min when all observations belong to a single class

↳ Find a tree with minimal impurity over all splits of all nodes

One can measure the quality of a split below node t:

$$\Delta i(S,t) = i(t) - \pi(l)\, i(l) - \pi(r)\, i(r)$$

Resubstitution Error:

$$i(t) = 1 - \max_j p(j|t)$$

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Decision Trees

~~Misclassification Error:~~

Gini impurity / Gini Index: ↱ frequency of cases of class A in node t

$$i(t) = p(A|t) * (1 - p(A|t))$$

↳ $$i(t) = \sum_j p(j|t) * (1 - p(j|t))$$

Entropy / Information Gain:

$$i(t) = -\sum_j p(j|t) \log_j p(j|t)$$

↳ Measure of the average amount of information generated by drawing ~~at~~ randomly with p replacement from this node.

If a node is pure, the class label yields zero information gain

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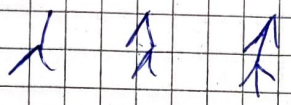
Random Forest

Features:
- → Popular classification method
- → runs efficiently
- → gives information of which variables are important in the classification
- → unbiased estimate of the generalization error during forest building (out-of-bag error rate)
- → robust to overfitting
- → robust to unbalanced class sizes

Multiple Trees: 

Feature Selection:

I) Preliminary ranking and elimination
- → Sort variables in decreasing order of RF scores of importance (Gini Index)
- → cancel the var's of small importance, in number of remaining var's

II) Variable selection

For interpretation:
- → construct the nested collection of RF models involving the k first variables, $k \in [1, m]$;
- → select the var's leading to the smallest OOB error

For prediction:
- → Starting from the ordered vars retained for interpretation, construct an ascending sequence of RF models by increasing and testing the variables stepwise.
- → the var's of the last model are selected

Data:
Land, Drunk - Victory
⋮

Resample:
Land, Drunk - Victory
!

Out of Bag
Quand, Ne - Drunk = Land ✗ Victory ✓

→ Tree → compute $OOB = \frac{\#\text{false predicted}}{\#\text{out of bag}}$

OOB, the average error for each calculated using predictions from the trees that do not contain in their respective bootstrap sample

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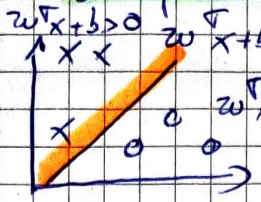
Support - Vector Machine (SVM)

Linear Separators: Binary classification



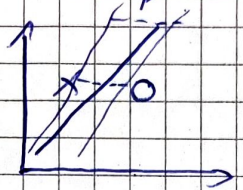
$f(x) = \text{sign}(w^T x + b)$

→ which separator is optimal

→ distance from example x_i to the separator is $r = \frac{w^T x_i + b}{\|w\|}$

→ examples closest to the hyperplane are support vectors

→ Margin ρ of the separator is the distance between support vectors



⇒ Maximum Margin Classification

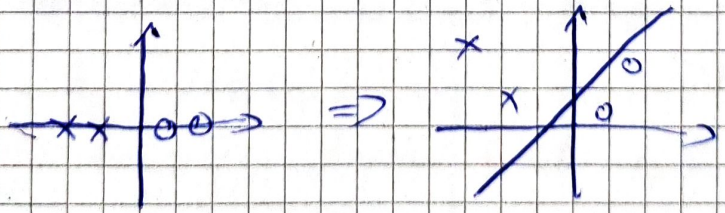
↳ good accuracy for PAC (probably approximately Correct)

↳ only support vectors matter; other training examples are ignorable!

Non-linear SVM & Feature Space

→ if not separable

→ Original feature space can be mapped to some higher-dimensional feature space where the training set is separable



1) Data in Machine Learning

a) Overfitting: Model perfectly explains training data, but performs poorly on new data

b)

Cross-Validation leaves a certain amount of data out of the training data and tests afterward if the model can correctly predict the test data

Helps Overfitting:

→ by providing a more reliable estimate of models ability to generalize to new data

→ by using multiple, non-overlapping subsets of the data for training and testing → wider range of data

c) Example why leave One Out

if we have a relatively small dataset, and we want to ensure that we are not overfitting to the limited number of observations we have. (eg: patients in disease research)

↳ helps to identify potential issues with overfitting and ensure that our model is robust enough

d)

Bootstrapping	Cross Validation
• random sampling with replacement	• partitioning of the data into folds
• used for estimating the sampling distribution of a statistic or making inferences about a population	• used for evaluation of the performance of a predictive model

CSB –

② Random Forest

a) follow tree based on entry data

b) feature = variable = node inside tree

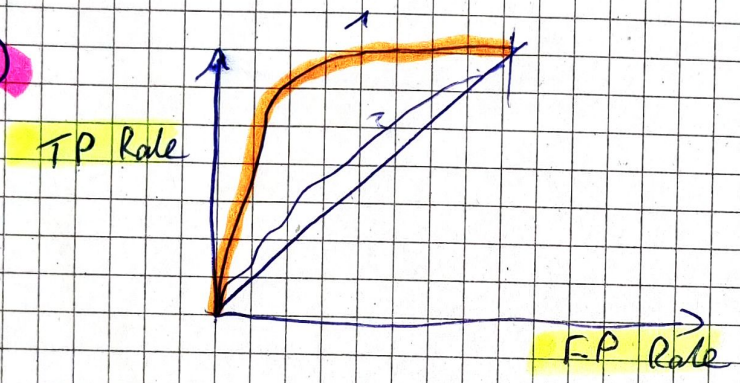
③

a) In Random forest, you can generate hundreds of trees and then aggregate the results of these trees.

↳ Individual tree is built on a subset of the features

↳ Individual tree is built on a subset of observations

④



1 is better ☺ TP rate is 100 %, when FP rate is only 25%

all positive samples are predicted positive ✓

↳ 2 has a FP rate 25%, only 37% TP

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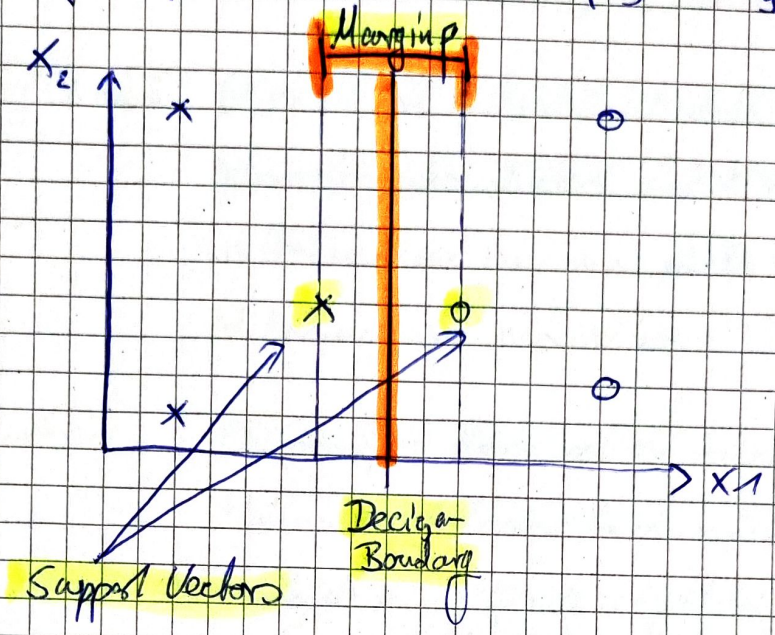
upper cases adhere

(4) SVM

An SVM can be called linear / non-linear model

Class 1 : $\begin{pmatrix} 1 & 9 \\ 5 & 5 \\ 1 & 2 \end{pmatrix}$ Class 2 : $\begin{pmatrix} 9 & 9 \\ 13 & 1 \\ 13 & 9 \end{pmatrix}$



CSB 4

Clustering

Unsupervised learning : no labels

eg Supervised
Classificatio
Regressi
↳ Logis
↳ SVM

Clusters : Groups of data objects that are more similar to each other than to objects from other groups

Clustering Goal : Given a set of unlabeled data objects, group them into clusters such that the intra-cluster differences are minimized and the inter-cluster differences are maximized

Why clustering ? Clustering a data set can reveal hidden relationships between the observations and features
↳ Objects in one group → highly similar
different groups → highly dissimilar

Methods : Density-based, Graph-based, Hierarchical, Model-based, Prototype-based (eg: k-means)

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K-Means

→ a partitional clustering algorithm

→ partitions the data into k clusters fixed a priori

I) Pick k, the number of clusters that you want to generate

II) Select k data points as cluster centers (centroids)

↳ Set of centroids $C = c_1, \dots, c_k$

III) Measure the distance between each point and the centroids (eg: Euclidean Distance) $\arg\min_{c_i \in C} dist(c_i, x)^2$

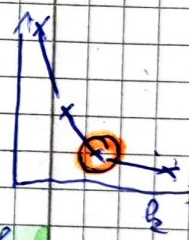
IV) Each data point is assigned to the cluster with the closest centroid

V) Update the centroids, with mean of all data points assigned to the centroid: $c_i = \frac{1}{|S_i|} \sum_{x_i \in S_i} x_i$

Repeat until convergence

→ If k is unknown?

Elbow Method: perform algorithm with different values of k



At each evaluation the Sum of Squared Error (SSE) is computed and then plotted: best value of k is the elbow point

SSE: Sum of the squared distances between centroid and each member of that cluster

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K-Means

Pros:

(+) Easy implementation & interpretation

(+) Fast for low dimensional data $O(tkn)$ (k: # clusters, n: # data points, t: # iterations)

Cons:

(-) to reach convergence, may need to run multiple times with diff. st

(-) lack consistency

(-) sensitive to noise and outliers

(-) # clusters must be known

(-) no guaranteed global convergence

(-) not suitable for complex data structures

(-) problematic if objects are not described by feature vectors

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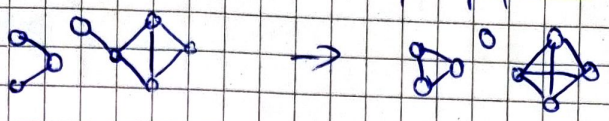
TransClust

- a clustering tool for biomedical datasets
- based on Weighted Transitive Graph Projection



- Transforms a given graph into a disjoint union of cliques by using a minimum number of edge modifications

In: → Set of objects M
→ Similarity Function $S: \binom{M}{2} \to \mathbb{R}$

Goal: → Find subsets (clusters) which are highly connected according to the similarity → eg find cluster of homologous proteins

- Similarity graph G; Nodes = Proteins, Edges = similarities with weights
- Edge weight threshold t to construct graph G'

  Existing edges: $w > t$

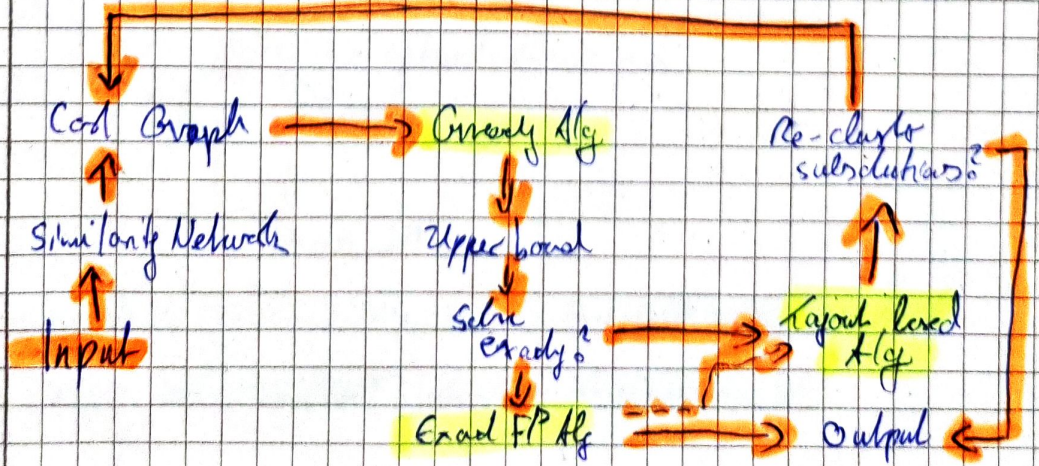
  Not existing edges: $w < t$

  Costs function for addel of edge: $c = |w - t|$

  ↖ modified Similarity Graph → $w' = w - t$ ↳ cost graph $w'' = |w'|$

- G' → a transitive graph G^* with minimal costs c

- → NP-Hard Problem
- → TransClust uses: Greedy Alg (fast but not guarantee), Exact Fixed-Parameter Alg (guarantee exact solution but exponential), Greedy Layout based Alg (need exact in polynomial time)



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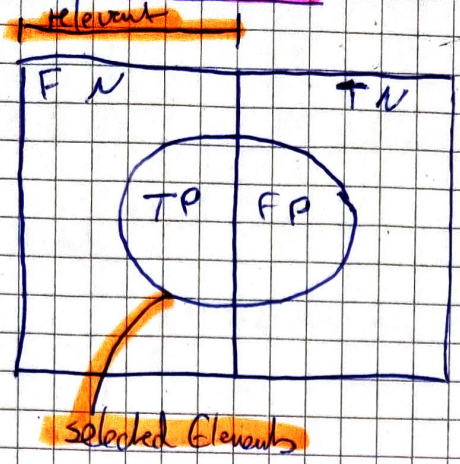
Evaluation: F-measure



How many selected items are relevant: Precision: $\frac{TP}{TP+FP}$

How many relevant items are selected: Recall: $\frac{TP}{TP+FN}$

→ Harmonic Mean: $F = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \in [0,1]$

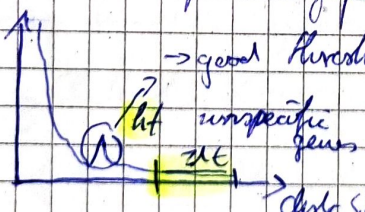
↳ the higher the better the clusters (predicted groups) match a given gold standard (known groups)

↳ can also be used for supervised classifiers.

→ How to find the right density parameter with gold standard?

Quality Measures is derived based on the gold standard

→ How to find the right density parameter without gold standard?

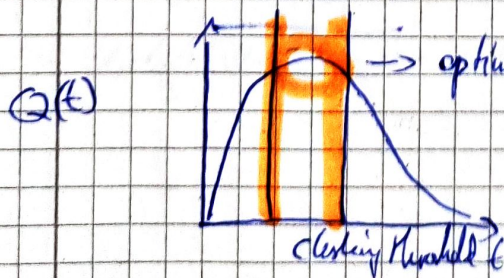


$Q(\varepsilon) = 2 \frac{\varepsilon_1 (1-\varepsilon_2)}{\varepsilon_1 + (1-\varepsilon_2)}$

ε_1 is peak

ε_2 ⟶

↳ Quality measure for each clustering threshold ε



→ Speeding up precomputations, Clusters with missing values

↳ Imputing similarity values for the threshold ε

(1) K-Means

$x1 = (1,2)$, $x2 = (5,8)$, $x3 = (7,3)$,
$x4 = (6,4)$, $x5 = (4,7)$

Distance Matrix

	x1	x2	x3	x4	x5
x1	0				
x2		0			
x3			0		
x4				0	
x5					0

a) K-means: x1, x3, x4 are centroids:

Cluster 1: x1,
Cluster 2: x5,
Cluster 3: x4, x2, x5

b)

$C_1' = (1,2)$

$C_2' = (7,3)$

$C_3' = \left(\frac{1}{3}\cdot(5+6+4), \frac{1}{3}\cdot(8+4+7)\right) = (5, 6\overline{33})$

$C_i = \frac{1}{|S_i|} \sum_{x_i \in S_i} x_i$

c) Continue K-means after how many iterations stop

It 2: Cluster 1: x1 , Cluster 2: x3, x4 , Cluster 3: x4 x5

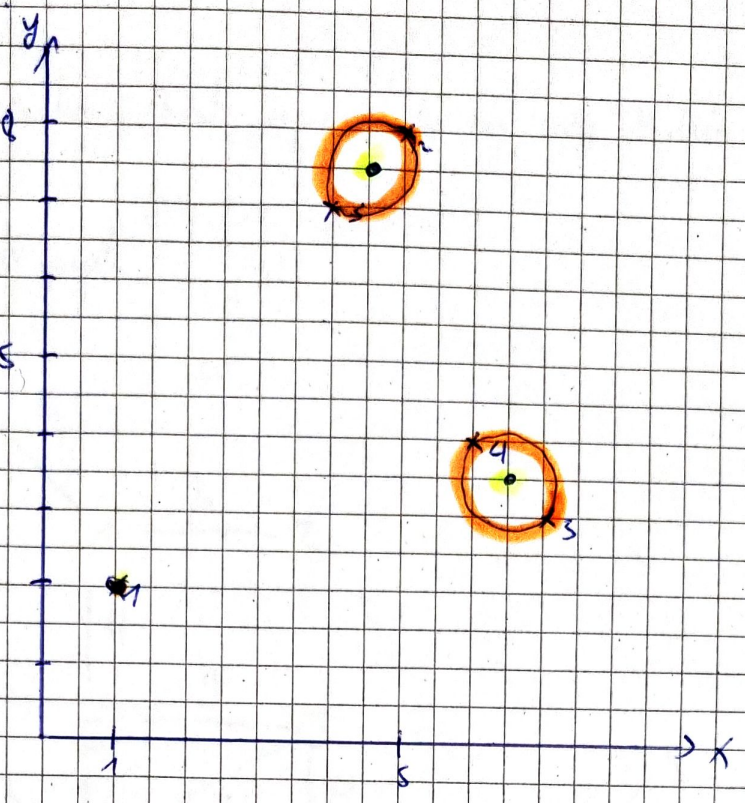
$C_1 = C_1 old$, $C_2 = (6,5, 3,5)$, $C_3 = (4,5, 7,5)$

It 3: Same → STOP

Update distances for centroids!

(10)

d)



e)

K-Means works best for clustering a convex shape.
To estimate the # of clusters without gold standard - one may use the Elbow Method.
Given a gold standard, one can use the F-measure to evaluate the clustering results, where 1 is best and 0 is worst.

f)

- K-means is very sensitive to the initial seed! 😕
- Bad initialization can lead to poor convergence (eg. outlier! 😕

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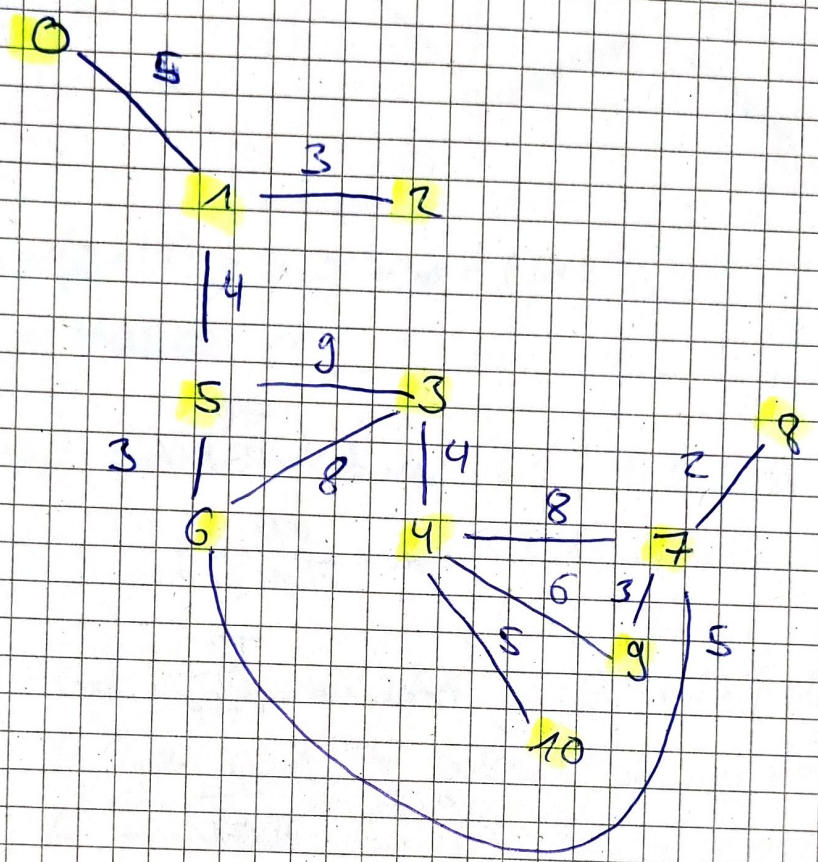
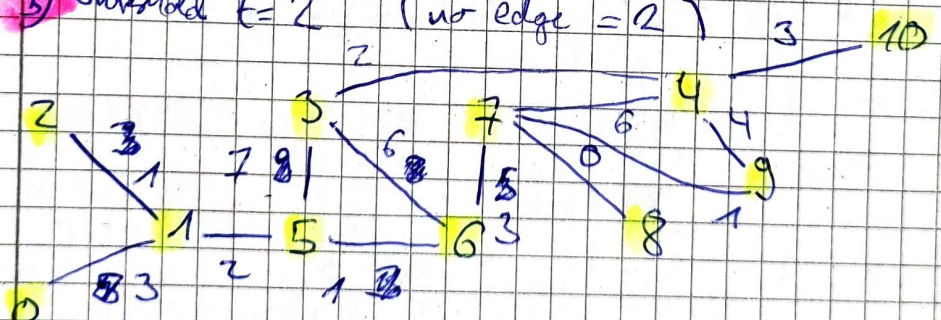
(7) Two Club

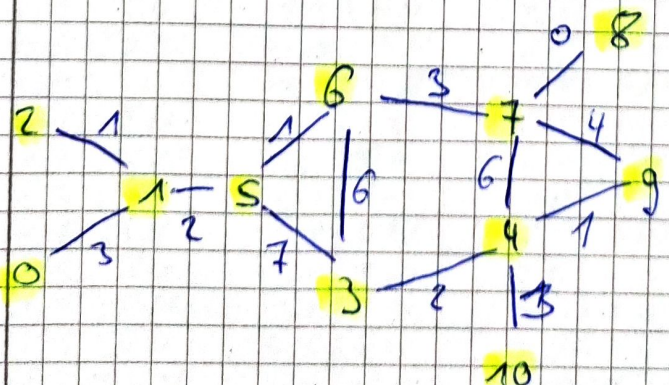
Weighted similarity Graph represent by similarity Matrix

a) Similarity graph

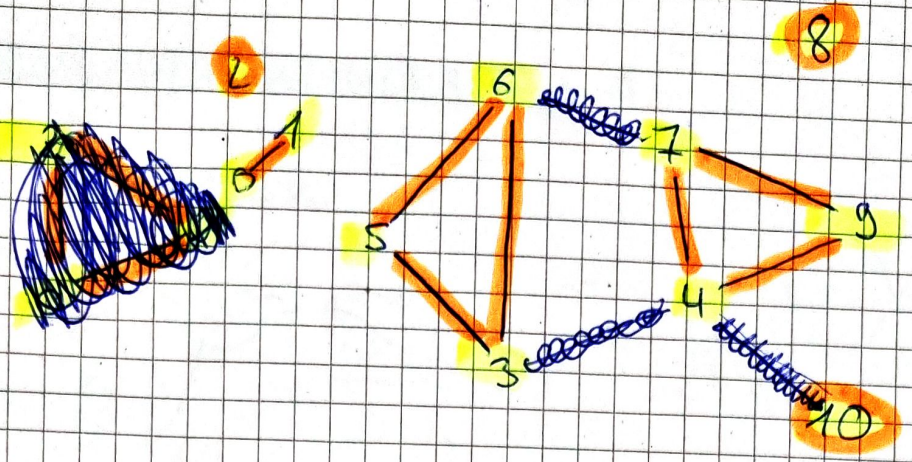


b) threshold t=2 (no edge = 2)





c)



d)

Costs = (2+1) + 8 0 + 3 + (3+2)

= 11

e) Given Gold Standard: C1: 1, 3, 6, 8, 0

C2: 2, 9, 16

C3: 4, 5, 7.

TN: Not in the same cluster in GS and in predicted

TP: In same cluster in GS and in predicted

FN: In same cluster in GS but not in predicted

FP: Not in the same cluster in GS but in predicted

0,1: TP	2,3: TN	4,9: FP
0,2: TN	2,4: TN	4,10: TN
0,3: FN	2,5: TN	5,6: FP
0,4: TN	2,6: TN	5,7: FN
0,5: TN	2,7: TN	5,8: TN
0,6: FN	2,8: TN	5,9: TN
0,7: TN	2,9: FN	5,10: TN
0,8: TN	2,10: FN	6,7: TN
0,9: FN	3,4: TN	6,8: FN
0,10: TN	3,5: FP	6,9: TN
1,2: TN	3,6: FP	6,10: TN
1,3: FN	3,7: TN	7,8: TN
1,4: TN	3,8: FN	7,9: FP
1,5: TN	3,9: TN	7,10: TN
1,6: FN	3,10: TN	8,9: TN
1,7: TN	4,5: FN	8,10: TN
1,8: FN	4,6: TN	9,10: FN
1,9: TN	4,7: TP	
1,10: TN	4,8: TN	

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{recall} = \frac{TP}{TP + FN}$$

$$F = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

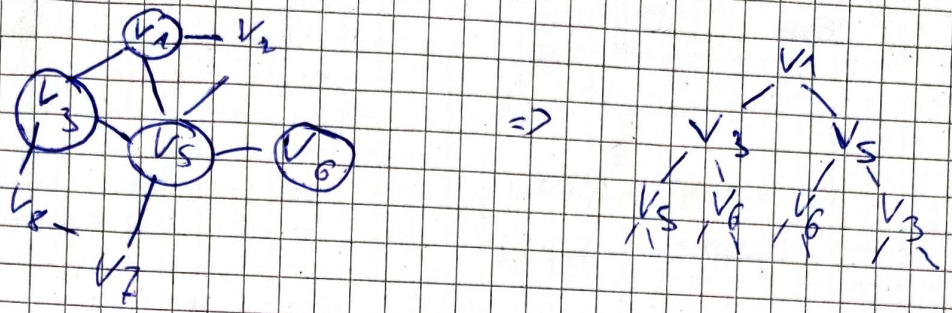
$$= 0,0133$$

↳ done

CSB -25

GraphRF - Supervised Analysis

Main Idea of Graphel-Based Random Forest



An ensemble learning method for discovering disease-associated modules from genomic profiling data.

=> ensemble learning method based on the random forest algorithm

=> Random Forest:

↳ combines the output of multiple decision trees to reach a single result

↳ Advantage over decision tree: corrects and prevents overfitting by using multiple trees

GraphRF

↳ Random Forest + Network => New Network

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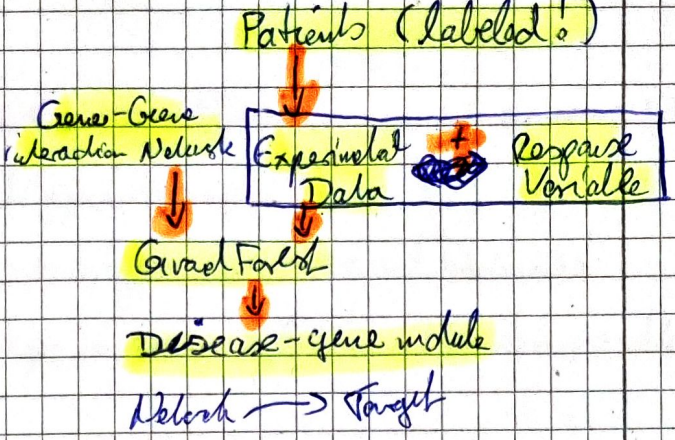
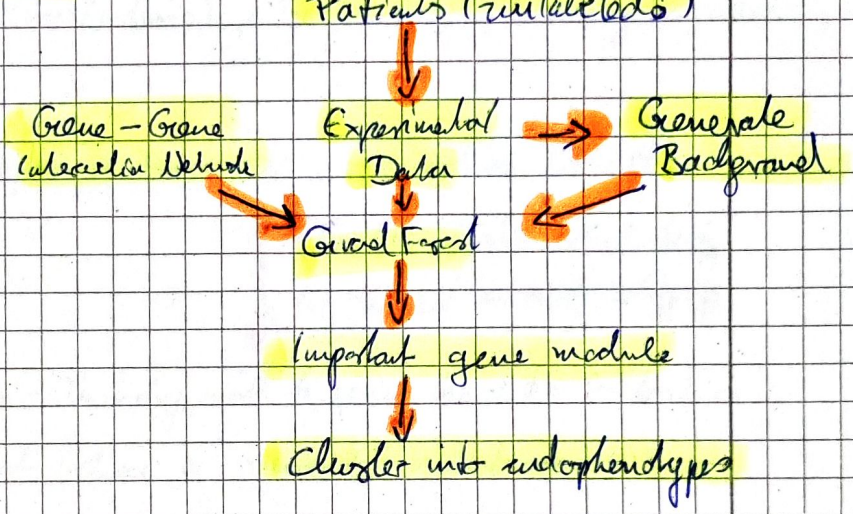
GiveadFast

Supervised Analysis:



Unsupervised Analysis:



→ upload expression data unlabeled

→ clusters data set recursively on computed pathways

↳ De novo Endophenotypes

→ also allows for survival analysis

→ Computes Kaplan-Meier plots and pvalues

→

1) Generate BGI from Input
2) Concatenate into style data set
3) Train GF model to predict FGI from BGI
4) Cluster input only on genes in module

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Network-constraint bi-clustering for de novo endophenotypes

BiGAnts / C-BiCoN:

Motivation:
- -> supervised methods are biased to current disease definitions
- -> Disease subtypes may exist
- -> Different mechanotypes require different treatments
- -> Shelved drugs might be effective and/or repurposable

Canal approaches not integrative enough:

Clustering (rows/columns), Bi-clustering, Community Detection

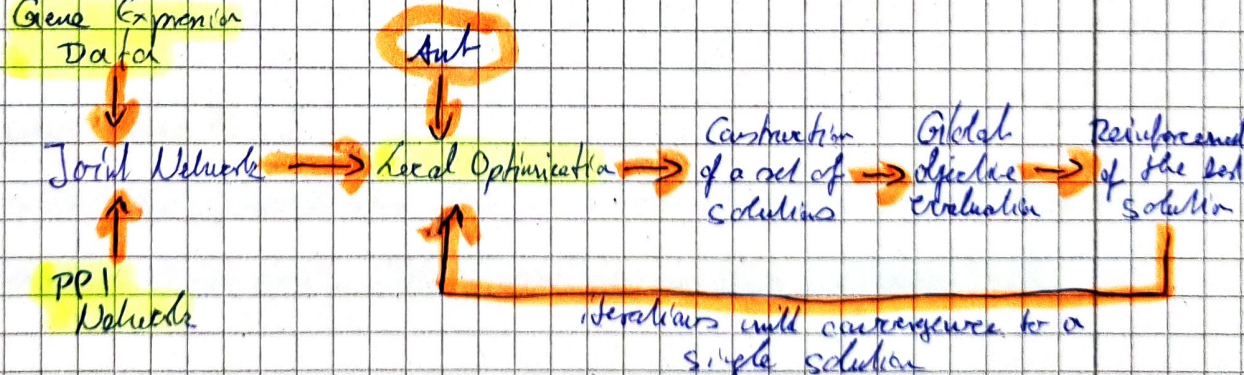
Idea:

Input: Expression Data & Interaction Network

↓ Network constraint bi-clustering for de novo mechanotypes

Output: Bi-Clustering with k genes inside the biclusters that induce a connected subnetwork

- -> restriction of genes in the biclusters to be functionally related & connected in Network
- -> Maximizes the expression differences of these two genes sets between two subgroups of patients



-> used to:

identify potential subgroups and biomarkers such as disease-associated genes

-> robustness to noise: considers only connected subnetworks

CSB

(1) Grand Forest

a) What is de novo endophenotyping?

de novo ≙ ~~new~~ new

phenotype = the set of observable characteristics or traits of an organism, eg: hair colour

endophenotype = measurable traits that are thought to be more closely linked to the genetic factors underlying a disorder.

de novo endophenotyping ≙ refers to the process of identifying and characterizing newly recognized or previously unknown measurable traits that are believed to be intermediate markers of the underlying genetic factors of a disorder

b)

Supervised analysis: input features and corresponding output labels (labeled data)

unsupervised analysis: the model has to figure out the labels (unlabeled data)

c)

Kaplan-Meier Plot:

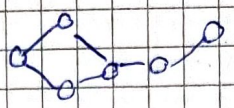
a graphical representation that visualizes the probability of an event occuring over time. Often used to analyse survival times.

y-axis: probability of the event of interest not occuring

x-axis: time passed

CSB

① BiCoN, a network-constrained biclustering approach



Extracted Subnetworks:

We see a connected subnetwork with k genes inside the bi-cluster

Heatmap:

one can see the biclustering

Survival Analysis:

Kaplan-Meier plot shows the difference in survival rate between the bi-clusters

What parameters uses BiCon:

→ minimal & maximal # of genes wanted in each subnetwork

→ gene set size, input size

→ max # of iterations

→ # of ants, can explore larger search space → increase runtime

→ evaporation rate, decrease the time until convergence